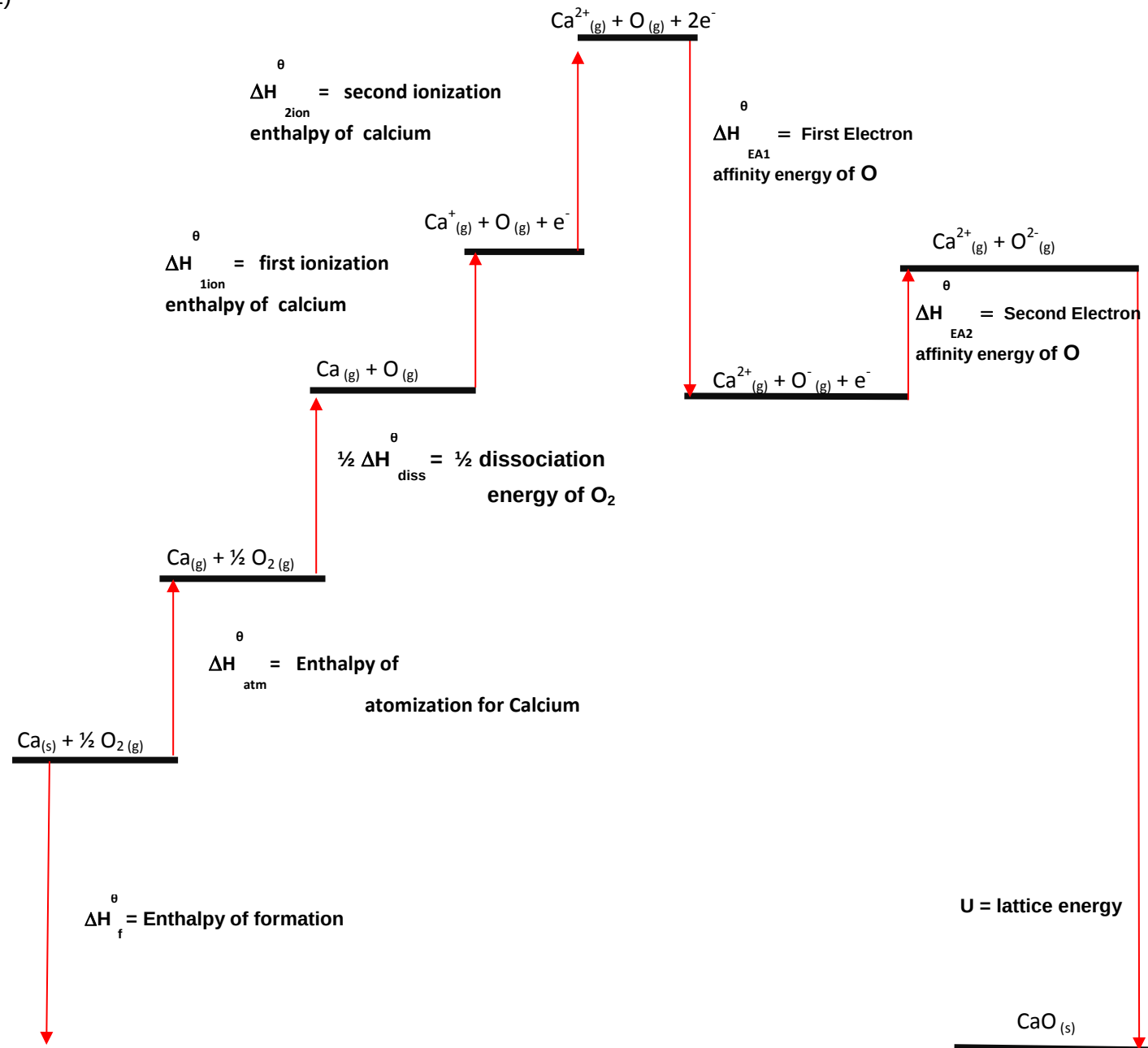


EXAM

Inorganic Chemistry

1)



2) Use the most appropriate definition to identify acids and bases in the following reactions (please, pay attention to the specific reaction environment, 10pt)

- a) Na_2O is a base according to the Lux-Flood definition
- b) HNO_3 is a base according to the solvent-system definition or is a base according to the Lux-Flood definition
- c) Fe^{2+} is an acid according to the Lewis acid definition
- d) PH_3 is a base according to the Lewis acid definition
- e) In this system urea is an acid according to the solvent-system definition

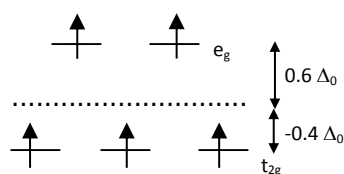
3)

Central atom	Ox. state	Coordination n.	Possible geometries
f) Rh	3+	6	Octahedral
g) Cd	2+	4	tetrahedral, square planar

4) Mn_3O_4 belongs to the spinel family: $(\text{A}^{2+})(\text{B}^{3+})_2\text{O}_4$, therefore oxidation number of manganese are 2+ and 3+.

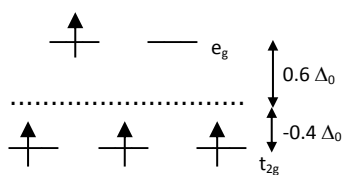
Oxygen is a weak field ligand (high spin complexes)

Mn^{2+} has 5 electrons in its 3d orbitals



$$\text{Crystal Field Stabilization Energy (CFSE)} = 3 \times (-0.4\Delta_0) + 2 \times (0.6\Delta_0) = 0$$

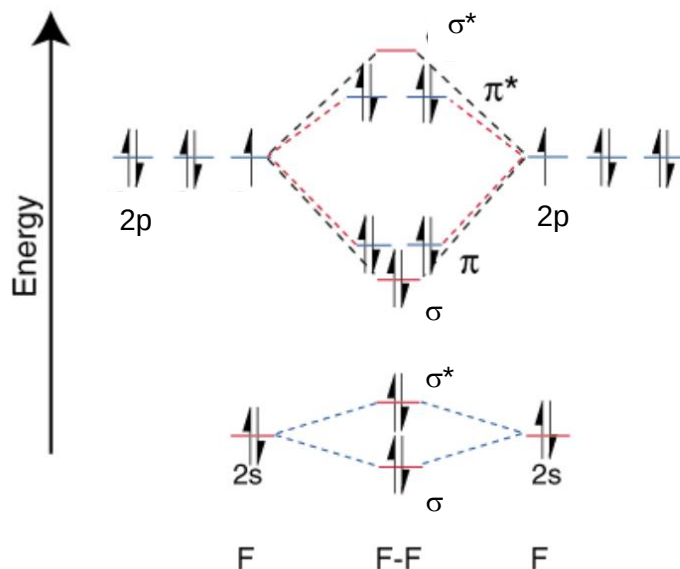
Mn^{3+} has 4 electrons in its 3d orbitals



$$\text{Crystal Field Stabilization Energy (CFSE)} = 3 \times (-0.4\Delta_0) + 1 \times (0.6\Delta_0) = -0.6\Delta_0$$

The stabilization of the octahedral field is higher for Mn^{3+} than for Mn^{2+} , which means that the most stable structure has most likely Mn^{3+} ions in the octahedral cavities, therefore we can predict a NORMAL spinel structure for this compound.

5)



Bond order = (electrons in bonding orbitals - electrons in antibonding orbitals)/2 = (8 - 6)/2 = 1
Bond order = 1, that is, a single bond.

6) Pauling defined electronegativity as the power of an atom in a molecule to attract shared electrons to itself and built an empirical scale.

This relative attraction for bonding electron pairs must reflect the comparative Z_{eff} . According to this assumption, Allred- Rochow defined electronegativity as:

$$\chi^{AR} = 0.359 \frac{Z^*}{r_{\text{cov}}^2} + 0.744 \text{ expressed in pm}$$

- 7) Slater proposed a formula that related the effective nuclear charge (Z_{eff}) to the atomic number (Z): $Z_{\text{eff}} = Z - \sigma$

where σ is a screening constant, which can be calculated according to empirical rules.

The main limitation of Slater's rules is that they are based on a simple model and the electrons in s and p orbitals give the same contribution to the screening constant.

The electron configuration of Oxygen is $1s^2 2s^2 2p^4$.

According to Slater's rules $\sigma = 5 \times 0.35 + 2 \times 0.85 = 3.45$

$$Z_{\text{eff}} = Z - \sigma = 8 - 3.45 = 4.55.$$

- 8) Both covalent networks and metals are solid systems. However, in covalent networks valence electrons are shared between couple of atoms, while in metals the valence electrons are free to move through the bulk metal structure. Metallic materials typically consist of elements with low electronegativity, while element with high electronegativity, as C, Si, and O, can form covalent networks.

Examples of covalent networks are:

- The diamond where each carbon atom is connected to other four carbon atoms by single bonds in a tridimensional structure.
- Silica glass which consist of a tridimensional and disordered network of silicon and oxygen atoms.